

Quick Check

Lesson 4: Cash from Waste: Categorising Plant Excretory Products and Their Value in Kenya

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.3: Excretion in Plants | Lesson 4

During the field walk, a learner crushes a lemongrass leaf and smells a strong scent. Which type of plant excretory product is mainly responsible for this smell?

A Gums and resins

B Essential oils

C Latex

D Tannins

Which row correctly matches the plant excretory product with its type and plant source?

A Gum arabic - alkaloid - Acacia senegal

B Papain - essential oil - papaya

C Rubber - latex - Hevea

D Quinine - tannin - Cinchona

A learner says, 'Latex from a rubber tree is the blood of the plant.' Which statement correctly describes latex?

- A** It is a fluid that carries oxygen around the plant
- B** It is a milky fluid stored in special cells and vessels, and it is not blood
- C** It is sugar solution made in the leaves and sent to the roots
- D** It is water and minerals moving up from the roots

Complete the CER argument. Claim: Plant excretory products are valuable in Kenyan markets. Evidence: quinine from Cinchona bark, rubber from Hevea latex and gum arabic from Acacia. Which is the best reasoning?

- A** The products are valuable because they are the main food that plants make for growth
- B** The products are valuable because they come only from plants that grow in Kenya
- C** Plants store these products in tissues such as bark and latex vessels, and people use their chemical properties, such as treating malaria, sealing and sticking
- D** The products are valuable because plants throw them away, so they are free and easy to collect

A group wants to move the DQB question 'Why is the sticky sap from the Acacia tree worth money?' to 'answered'. Which evidence from today best justifies the move?

- A** The sap oozes out only when the tree is cut, so it has no use
- B** Acacia sap hardens into gum arabic, which collectors in Turkana and Marsabit sell, so a plant waste earns money
- C** All liquids that come out of plants are sold in markets
- D** Acacia trees have thorns and grow in dry areas of Kenya

A group is updating its class model of plant excretion. Which annotation is correct?

- A** Latex is made in the leaf glands
- B** Gums and resins ooze out of stem wounds
- C** Essential oils are stored in latex vessels in the stem
- D** Tannins are found only in the root hairs

Which statement correctly contrasts guttation and transpiration in plants?

- A** Both are the loss of latex from cut stems, but they happen at different times of day
- B** Guttation is loss of liquid water droplets from leaf edges (often on humid mornings), while transpiration is loss of water vapour, mainly through stomata
- C** Guttation is loss of water vapour through stomata, while transpiration is loss of liquid droplets from leaf edges
- D** Guttation happens only in old yellow leaves, while transpiration happens only in young leaves

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).